



Eigenvalues and Eigenvectors

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Recap

- Matrices
- Vectors
- Basis and unit vectors
- Linear combination of vectors
- Linear Transformations
- Change of Basis $\rightarrow$ ref. $\rightarrow$ new coordinate
 $\rightarrow$ Linear transformation

Change of Basis

Eigenvectors and Eigenvalues

Suppose we have a coordinate system and we translate it to a new coordinate space.

- Majority of the vectors $\vec{V}$ from the original space will be knocked off from their span.
- However, there exists some special set of vectors which remain on their span and are scaled by some factor.

These set of vectors are known as Eigenvectors.

The scaling of the eigenvectors is determined by Eigenvalues.

E.X.

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$$

→ $\vec{V} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ ✓
 ↳ factor of 2

Eigenvectors

Eigenvalue → 2

$$\begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

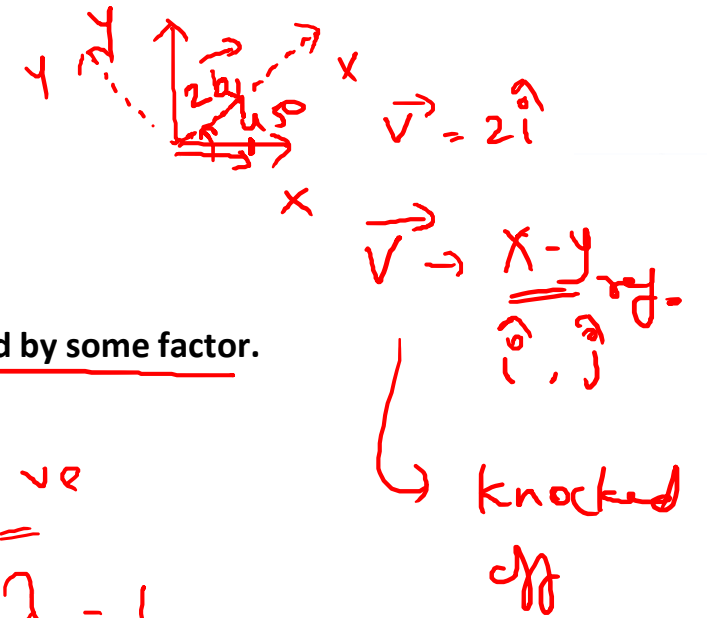
$$\lambda = 1 \rightarrow \lambda > 1 \rightarrow$$

↳ $0 < \lambda < 1 \rightarrow$ compressor

$$\lambda < 0 \quad \lambda < -1$$

$$\vec{V} = \begin{bmatrix} 3 \\ -1 \end{bmatrix} \rightarrow$$

~~$$\begin{bmatrix} 3 \\ -1 \end{bmatrix}$$~~



Mathematical Definition

Mathematically eigenvectors and eigenvalues can be obtained from:

$$A\vec{V} = \lambda\vec{V}$$

Where,

A is the transformation matrix

$\vec{V}$ is the eigenvector

λ is the eigenvalue



2×2
 $\text{matrix} \times \text{vector} = \text{Scalar} \times \text{vector}$

$$\lambda [\vec{V}]$$

$$\lambda \vec{V} = \lambda I \vec{V}$$

$$A\vec{V} = \lambda I \vec{V}$$

$$A\vec{V} - \lambda I \vec{V} = 0 \rightarrow \vec{V} = 0$$

$$\text{or } (A - \lambda I) \vec{V} = 0$$

Change of basis

Linear tran.

$\rightarrow$ origin $\rightarrow$ remains on it's path

$$(A - \lambda I) \vec{V} = 0$$

$\rightarrow$ non-zero

$$\rightarrow A - \lambda I = 0$$

$\lambda \rightarrow$ Eigenvalues

Eigenvectors and Eigenvalues

E.X.

$$A = \begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}$$

$$A\vec{v} = \lambda I\vec{v}$$

$$\det(A - \lambda I) = 0$$

$\lambda \rightarrow$ eigenvalues

$$\begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = 0$$

$$\det \begin{bmatrix} 2-\lambda & 2 \\ 1 & 3-\lambda \end{bmatrix} = 0$$

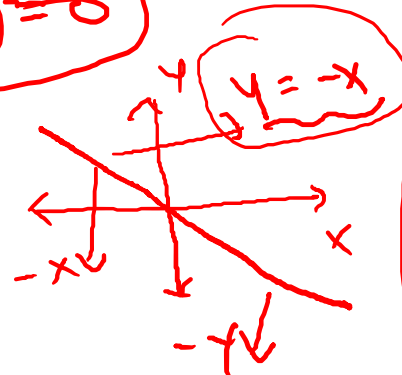
$$(2-\lambda)(3-\lambda) - (2)(1) = 0$$

$$\lambda^2 - 5\lambda + 4 = 0$$

$$\lambda = 1 \text{ \& } \lambda = 4$$

$$\det(A - \lambda I) = 0$$

Set $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$



(E.X.)

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$$

$$\lambda = 2 \text{ or } \lambda = 3$$

Eigen Vectors

$$(A - \lambda I)\vec{v} = 0$$

$$\begin{bmatrix} 3-2 & 1 \\ 0 & 2-2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{bmatrix} x+y \\ 0 \end{bmatrix} = 0$$

$$x+y = 0$$

$$y = -x$$

Eigenvectors and Eigenvalues

E.X.

$$A = \begin{bmatrix} 2 & 0 & 0 \\ -1 & 1 & 0 \\ 5 & 3 & -3 \end{bmatrix}$$

$\det(A - \lambda I) = 0$ $\uparrow$ 3×3

$$\begin{bmatrix} 2 & 0 & 0 \\ -1 & 1 & 0 \\ 5 & 3 & -3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\det \begin{bmatrix} 2-\lambda & 0 & 0 \\ -1 & 1-\lambda & 0 \\ 5 & 3 & -3-\lambda \end{bmatrix} = 0$$

$$(2-\lambda) \left[(1-\lambda)(-3-\lambda) - (3)(0) \right]$$

$$(2-\lambda)(1-\lambda)(-3-\lambda) = 0$$

$$\begin{aligned} \lambda &= 1 \\ \lambda &= 2 \\ \lambda &= -3 \end{aligned}$$

$$\begin{aligned} -3-\lambda &= 0 \\ -3 &= \lambda \end{aligned}$$

3 eigenvalues
3 sets of eigenvectors

$$(A - \lambda I) \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

~~Eigenvectors~~

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$\lambda = \sqrt{-1}$$

$$\begin{matrix} i & | & j \\ \hline u & & v \end{matrix}$$

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$\lambda = 0, \lambda = 2$$

$$A = [b_1 \ b_2]$$

$i \ j$

E.X.

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

$\vec{b}_1 \qquad \vec{b}_2$

$$y_2 \leftarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$\vec{x}$

eigen vectors

Eigenbasis

If The basis vectors
are eigenvectors

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix}$$

$\lambda = -1, \lambda = 2$

$A \rightarrow$ diagonal

$$A \sim \begin{pmatrix} -5 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1/3 \end{pmatrix}$$

λ

Series of term $\rightarrow$ $\square \rightarrow$ entry

$$\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \cdot \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \cdot \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \cdots \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} =$$

$$\begin{bmatrix} 3^n & 0 \\ 0 & 2^n \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3^n x \\ 2^n y \end{bmatrix}$$



Thank you!